

Thesis of the end-of-study project
TO OBTAIN THE STATE ENGINEERING DIPLOMA
Major: Advanced Software Engineering for Digital Services

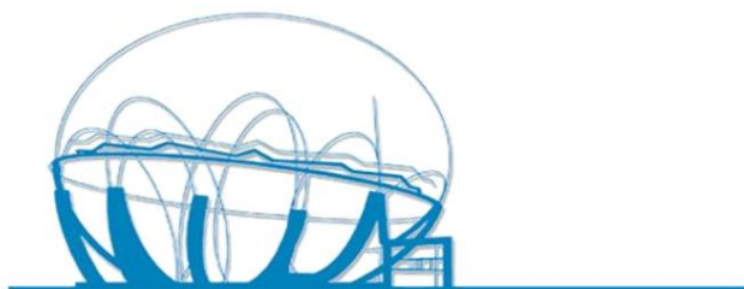
Project Title

Authored by :

Ms. Bouchra BOUDIOUANE

Defended on October XX, 2026, before the jury composed of :

Mr. Mostafa Bellafkih	INPT	Examiner
Mr. Nawfel Azami	INPT	Examiner
Mr. Khalid Soussi	INPT	Supervisor
Mr. Ajay Kumar	Oracle	Supervisor



AGENCE NATIONALE DE RÉGLEMENTATION DES TÉLÉCOMMUNICATIONS
INSTITUT NATIONAL DES POSTES ET TÉLÉCOMMUNICATIONS
Class of 2025–2026

Dedicated to

Words cannot express my gratitude for the countless sacrifices you made for my happiness, education, and well-being. Your unconditional love, support, patience, prayers, and encouragement have guided me through every challenge and achievement. You have always believed in me and inspired me to pursue my dreams with confidence. This accomplishment is as much yours as it is mine.

Friends and family (2nd paragraph)

Bouchra Boudiouane

Acknowledgments

Thank your supervisors (INPT and company), the managers and HR, the teachers at INPT and the jury.

Résumé

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Praesent nec dapibus justo. Donec sagittis vulputate ante sed porttitor. Suspendisse sit amet nisl massa. Curabitur nec nisl condimentum, egestas ex vitae, dapibus enim. Etiam iaculis, erat faucibus pellentesque sagittis, nisi justo sollicitudin nibh, et condimentum augue massa non turpis. Proin commodo enim fermentum suscipit condimentum. Maecenas molestie, dui nec vestibulum rhoncus, arcu nisl faucibus neque, a ornare nisi massa ac eros. Aenean id velit sit amet lacus mattis varius. Donec fringilla massa sed nisi eleifend, a aliquet mi tempus. Nunc posuere euismod est, nec tristique augue lobortis non. Sed sodales sem ut metus tempus ullamcorper.

Mots clés : xxx, xxx, xxx, xxx.

Summary

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Praesent nec dapibus justo. Donec sagittis vulputate ante sed porttitor. Suspendisse sit amet nisl massa. Curabitur nec nisl condimentum, egestas ex vitae, dapibus enim. Etiam iaculis, erat faucibus pellentesque sagittis, nisi justo sollicitudin nibh, et condimentum augue massa non turpis. Proin commodo enim fermentum suscipit condimentum. Maecenas molestie, dui nec vestibulum rhoncus, arcu nisl faucibus neque, a ornare nisi massa ac eros. Aenean id velit sit amet lacus mattis varius. Donec fringilla massa sed nisi eleifend, a aliquet mi tempus. Nunc posuere euismod est, nec tristique augue lobortis non. Sed sodales sem ut metus tempus ullamcorper.

Key Words : xxxx, xxxx, xxx, xxx.

ملخص

ملخص بالعربية، تأكد من أن الترجمة منطقية.

الكلمات المفتاحية:

مثال

List of Figures

1.1	Oracle logo	1
1.2	Location of Oracle MADC	2
1.3	Major milestones of Oracle MADC	3
1.4	Oracle Labs logo	3
2.1	Oracle entities	7
2.2	DIS Region Availability	7
2.3	Data intelligence logo	8
2.4	GBU Analytics and Machine Learning Architecture	9
2.5	DIS Microservices Architecture	11
2.6	Auth Service Sequence Diagram	12
3.1	OAC provisioning in the primary and standby regions	14
3.2	Simplified architecture of the OAC cross-region DR solution	16
3.3	Tenant-level OAC provisioning during a DR event	17
3.4	Switchback to the primary region and cleanup of standby resources	19

List of Tables

3.1	Functional requirements of the OAC DR solution	15
3.2	Principal validation scenarios for OAC cross-region DR	20
C.1	Table Example	28
C.2	Long table Example	28

Listings

C.1	Bash example	27
C.2	Python example	27

Contents

Dedication	i
Acknowledgments	ii
Résumé	iii
Summary	iv
Arabic Abstract	v
List of Figures	vi
List of Tables	vii
List of Listings	viii
Table of Contents	x
1 General Project Context	1
1.1 Host Organization	1
1.2 Oracle Corporation	1
1.2.1 History	2
1.2.2 Oracle MADC	2
1.2.3 Oracle Labs	3
1.2.4 Organizational Structure	3
1.2.5 Organizational Culture	4
2 Technical and Organizational Context	6
2.1 Organizational Context within Oracle	6
2.2 DIS Region Availability and Status	7
2.3 What Is Data Intelligence?	8
2.4 DIS Purpose and Role	8

2.5	DIS Architecture Overview	9
2.5.1	Data Ingestion	9
2.5.2	Data Processing	10
2.5.3	Visualization and Data Analysis	10
2.6	DIS Microservices Overview	10
2.6.1	Orchestration Service	11
2.6.2	Authentication Service	11
3	Proposed Solutions and Development	13
3.1	Introduction	13
3.2	OAC Cross-Region Disaster Recovery Automation	13
3.2.1	Overview of the Existing Recovery Context	13
3.2.2	Problem Statement	14
3.2.3	Solution Requirements	15
3.2.3.1	Functional Requirements	15
3.2.3.2	Non-Functional Requirements	15
3.2.4	Proposed Solution Design	16
3.2.4.1	General Architecture	16
3.2.4.2	Backup and Cross-Region Preparation	17
3.2.4.3	Disaster Recovery Workflow	17
3.2.4.4	Retry and Rollback Strategy	18
3.2.4.5	Switchback and Cleanup	18
3.2.5	Validation	19
3.2.6	Conclusion	20
4	Chapter 4 title	22
	General Conclusion and Perspectives	23
A	Glossary	24
B	Acronyms	25
C	Some subject you want to expand on	27
	Bibliography	29

Chapter 1

General Project Context

Introduction

This chapter presents the organization that hosted the end-of-study internship. It begins with an overview of Oracle Corporation and its development, then introduces the Morocco Advanced Development Center (MADC) and Oracle Labs. Finally, it describes Oracle's organizational structure and the principal values that shape its corporate culture.

1.1 Host Organization

Businesses, technology companies, and public-sector organizations depend on databases and cloud services to manage their infrastructure, along with core enterprise systems such as ERP, CRM, EPM, and SPM. These technologies are now essential to their daily operations.

Oracle is a leading U.S. multinational technology company specializing in databases, cloud services, and enterprise software. It develops and delivers these technologies to organizations worldwide while placing a strong emphasis on innovation, reliability, and customer satisfaction.

1.2 Oracle Corporation



Figure 1.1: Oracle logo

1.2.1 History

Oracle Corporation is one of the world's largest enterprise software and cloud-computing companies. Its history began in 1977, when Larry Ellison, Bob Miner, and Ed Oates founded Software Development Laboratories (SDL) in Santa Clara, California. Their objective was to commercialize the emerging concept of the relational database. The company developed one of the first commercial relational database management systems and launched Oracle Version 2 in 1979. It was renamed Oracle Systems Corporation in 1983 and subsequently became a major provider of database technology.

During the 1980s, Oracle experienced rapid growth and went public in 1986. Although the company faced a serious financial crisis in 1990 due to accounting issues, it recovered under Larry Ellison's leadership and expanded beyond databases into enterprise software.

In the 2000s, Oracle expanded through major acquisitions, including PeopleSoft, Siebel Systems, BEA Systems, and Sun Microsystems. These acquisitions brought technologies such as Java and MySQL into Oracle's portfolio. During the 2010s and 2020s, Oracle strengthened its focus on cloud computing through Oracle Cloud Infrastructure (OCI) and cloud-based business applications.

Today, Oracle is one of the world's largest technology companies, providing database systems, cloud infrastructure, enterprise applications, and AI services to organizations worldwide.

1.2.2 Oracle MADC



Figure 1.2: Location of Oracle MADC

The Morocco Advanced Development Center (MADC), established by Oracle Corporation in Casablanca, is Oracle's first R&D center in Africa. Opened in 2015, it is one of Oracle's principal development centers in the EMEA region. The center brings together software developers, engineers, researchers, and project managers who contribute to projects across several Oracle product lines.

Their work covers fields such as artificial intelligence, augmented, virtual, and mixed reality, big-data analytics, blockchain, cloud computing, cybersecurity, the Internet of Things, machine learning,

mobile computing, and serverless computing. MADC also supports graduate recruitment, internship programs, and research collaboration with local academic institutions. Its teams have contributed to Oracle Cloud Infrastructure (OCI), Autonomous Data Warehouse (ADW), and other Oracle products and services.

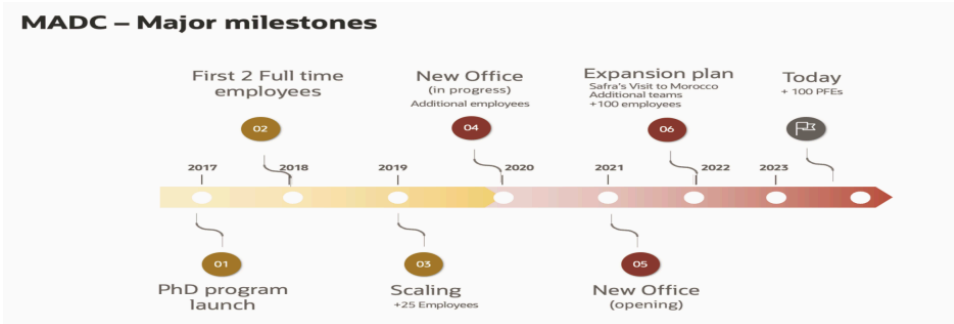


Figure 1.3: Major milestones of Oracle MADC

1.2.3 Oracle Labs

Oracle Labs, the company’s research organization, focuses on emerging technologies that may shape future Oracle products and services. Its researchers investigate innovative approaches and undertake projects involving substantial technical uncertainty or long-term research challenges. The objective is to transform research outcomes into practical technologies with measurable industrial impact. For example, work associated with Oracle’s research activities has contributed to areas such as the Java programming language and chip multithreading.

Oracle Labs plays a significant role in Oracle’s research and development efforts. Its diverse research portfolio helps Oracle identify problems, devise innovative solutions, and develop products that support future growth and innovation.



Figure 1.4: Oracle Labs logo

1.2.4 Organizational Structure

Oracle combines a hierarchical management structure with product-, service-, and region-oriented business units. The Board of Directors oversees the company’s strategic direction and performance, while the executive leadership team coordinates its operations. Functional departments such as sales,

marketing, finance, human resources, legal affairs, and information technology support the different business units.

Within this matrix structure, employees may collaborate across technical functions, products, and geographical regions. This organization supports coordinated decision-making while enabling multidisciplinary teams to work toward shared objectives.

1.2.5 Organizational Culture

Oracle has a strong corporate culture that prioritizes innovation, customer focus, and excellence. The company's history of innovation, emphasis on customer satisfaction, and commitment to fostering a supportive and collaborative work environment all shape its culture. Oracle's culture emphasizes the following values:

- **Innovation:** The company has a long history of creating products and technologies that have influenced the technology sector. This culture of innovation is supported by investment in research and development and by providing employees with the tools and encouragement needed to test new concepts and create new products.
- **Customer satisfaction:** Oracle's customer-centric approach reflects its commitment to providing products and services that meet its clients' needs. Employees are encouraged to collaborate closely with clients, understand their requirements, and develop suitable solutions.
- **Quality:** Oracle is committed to high-quality operations, products, and services. The company expects its employees to strive for excellence and provides them with the tools, resources, and support required to succeed.
- **Collaboration and teamwork:** Effective collaboration is essential to achieving Oracle's goals and delivering value to customers. The company encourages employees to work across departments and functions and provides tools and resources that facilitate communication and collaboration.

Overall, Oracle's organizational culture is characterized by a commitment to innovation, customer focus, excellence, collaboration, and teamwork. These values are reflected in the company's products, services, and operations and guide employees across its different organizational levels.

Conclusion

This chapter introduced Oracle Corporation and the environment in which the internship was conducted. It highlighted the role of Oracle MADDC as a development and research center, the

contribution of Oracle Labs to technological innovation, and the organizational principles that support collaboration across the company. This context provides the foundation for understanding the DIS platform and the technical environment presented in the next chapter.

Chapter 2

Technical and Organizational Context

Introduction

As Oracle continues to expand the adoption of the Data Intelligence Services (DIS) platform across its Global Business Units (GBUs), the choice of tools, technologies, and infrastructure becomes increasingly important. This chapter provides the technical foundation necessary to understand the rationale behind the design and implementation of the DIS platform. It introduces the problem domain, presents a high-level overview of the system, discusses key architectural considerations, and highlights the core technologies that underpin DIS.

Because this internship focused on identifying and addressing integration challenges encountered by GBUs, the chapter first explains their role within Oracle. It then presents the position of DIS within the organization, its regional availability, and its purpose as a shared analytics platform. This background establishes the context required to understand the work carried out during the internship.

2.1 Organizational Context within Oracle

Oracle Corporation is structured around two major pillars:

- The broader Oracle organization, which focuses primarily on infrastructure and platform services such as OCI.
- Global Business Units (GBUs), which deliver SaaS solutions tailored to specific industry domains such as healthcare, finance, hospitality, and retail.

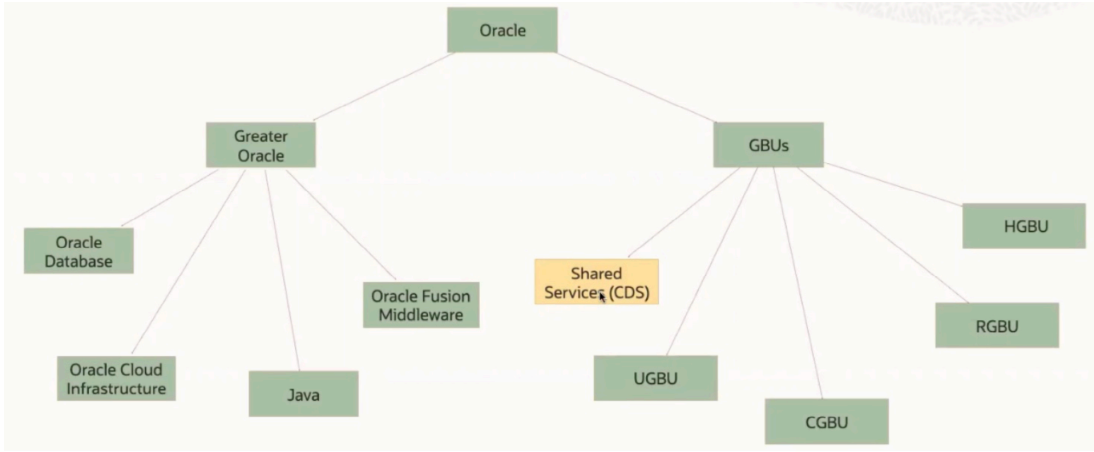


Figure 2.1: Oracle entities

To support these GBUs with shared tools and infrastructure, Oracle established Shared Services. This organization includes Cloud Development Services (CDS), which develops common platforms and capabilities used across GBUs. The Data Intelligence Services team is part of CDS and helps bridge the infrastructure and application domains.

DIS provides a managed machine-learning and analytics platform for use across Oracle’s GBUs. Instead of requiring each GBU to build and operate a separate data and AI platform, DIS offers centralized, reusable services. SaaS products can therefore integrate machine-learning and analytics capabilities without having to manage the underlying infrastructure independently. This shared approach improves consistency, reduces duplicated effort, and accelerates innovation across Oracle’s industry-specific SaaS offerings.

2.2 DIS Region Availability and Status

DIS is intended to be deployed in the OCI regions where the required supporting services are available. Deployment priorities are determined according to forecast demand from GBU partners and their regional requirements. Figure 2.2 presents the planned availability of DIS across the relevant regions.

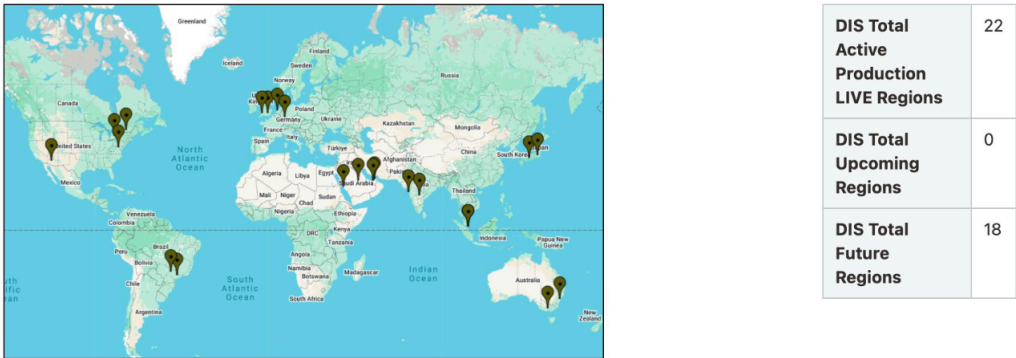


Figure 2.2: DIS Region Availability

2.3 What Is Data Intelligence?

Data intelligence refers to an organization's ability to collect, process, analyze, and interpret data in order to generate actionable insights. It combines tools, technologies, governance practices, and analytical methods that transform raw data into information that supports decision-making. By using their data assets effectively, organizations can improve operational efficiency, identify opportunities, and support strategic initiatives. In a data-driven environment, these capabilities are important for sustaining innovation and delivering value across business units.



Figure 2.3: Data intelligence logo

2.4 DIS Purpose and Role

DIS is an internal Oracle platform that provides product teams with a reusable analytics stack. Instead of requiring each GBU to build its analytics infrastructure from the ground up, DIS provides shared capabilities for data ingestion, storage, processing, machine learning, and visualization. A partner SaaS application can therefore integrate dashboards, reports, predictive models, and analytics pages through DIS rather than developing and operating each capability independently.

From a business perspective, Oracle's GBUs deliver SaaS solutions to their customers, while DIS provides the analytics layer embedded within those offerings. The platform provisions and manages services such as Oracle Analytics Server (OAS), Autonomous Data Warehouse (ADW), and Apache Spark, together with other analytics and data-processing components. End users access these capabilities through the corresponding SaaS applications.

By leveraging DIS for analytics-intensive workloads, partners benefit from the following advantages:

- Avoid building and maintaining a custom analytics technology stack.
- Gain access to a cost-effective and scalable architecture tailored to analytics requirements.
- Focus on developing advanced machine learning capabilities using built-in frameworks and libraries.

- Maintain full control and flexibility over data ingestion, storage, processing, and visualization.
- Reduce Cloud Service Security Approval Process (CSSAP) overhead by adopting DIS's pre-approved architectural patterns.

2.5 DIS Architecture Overview

Oracle Global Business Unit Analytics (GBUA), also known as the Data Intelligence Service (DIS), is a managed service platform focused on delivering scalable, cloud-native data analytics and machine learning capabilities for Oracle SaaS applications. It enables business units to integrate and leverage machine learning (ML)-based insights without managing the underlying infrastructure or analytics tooling.

The DIS platform architecture can be logically divided into three core layers, as shown in Figure 2.4.

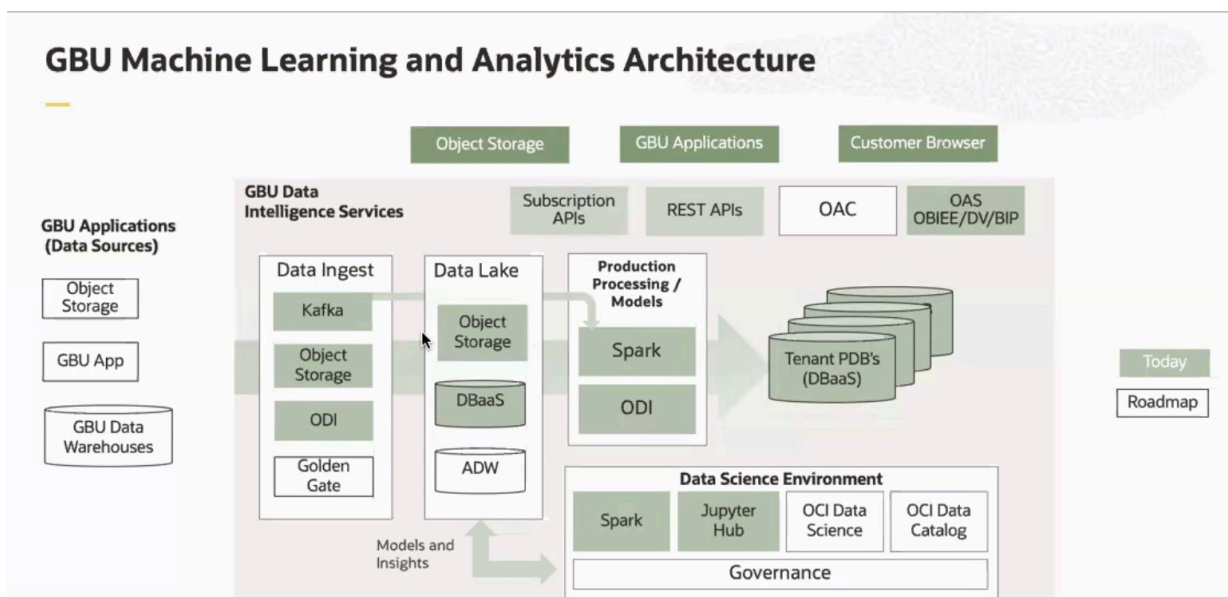


Figure 2.4: GBU Analytics and Machine Learning Architecture

2.5.1 Data Ingestion

The data ingestion layer supports multiple methods for collecting data from Oracle SaaS applications and partner systems:

- **REST Ingestion:** Data can be streamed into the platform using REST endpoints. These endpoints perform inline validation and Personally Identifiable Information (PII) detection during ingestion.

- **Oracle Data Integrator (ODI):** ODI is used to extract and load data from partner Database-as-a-Service (DBaaS) systems. It supports standard Extract, Transform, and Load (ETL) operations.
- **Direct Upload to Object Storage:** Large datasets, such as log files and historical data, can be uploaded directly to Oracle Object Storage for later processing.

2.5.2 Data Processing

The data processing layer provides distributed data processing using Apache Spark. It executes machine learning pipelines and data transformation jobs, storing the processed results in Oracle DBaaS. Spark jobs are tenant-isolated and are designed to periodically retrain machine learning models using historical data stored in Object Storage.

2.5.3 Visualization and Data Analysis

After data has been processed and stored, Oracle Business Intelligence Enterprise Edition (OBIEE) or Oracle Analytics Cloud (OAC) is used for report generation and dashboard visualization. Access is managed through tenant Identity Cloud Service (IDCS), and reports are customized according to user roles. OBIEE also provides self-service analytics capabilities, enabling administrative users with limited SQL or data modeling expertise to create reports and perform data analysis.

2.6 DIS Microservices Overview

The DIS platform implements a comprehensive microservices-based architecture in which loosely coupled services communicate through RESTful APIs. All services are containerized and deployed on the Oracle Kubernetes Engine (OKE) platform, enabling scalability, fault tolerance, and independent service deployment.

Figure 2.5 illustrates the high-level microservices architecture of the DIS platform.

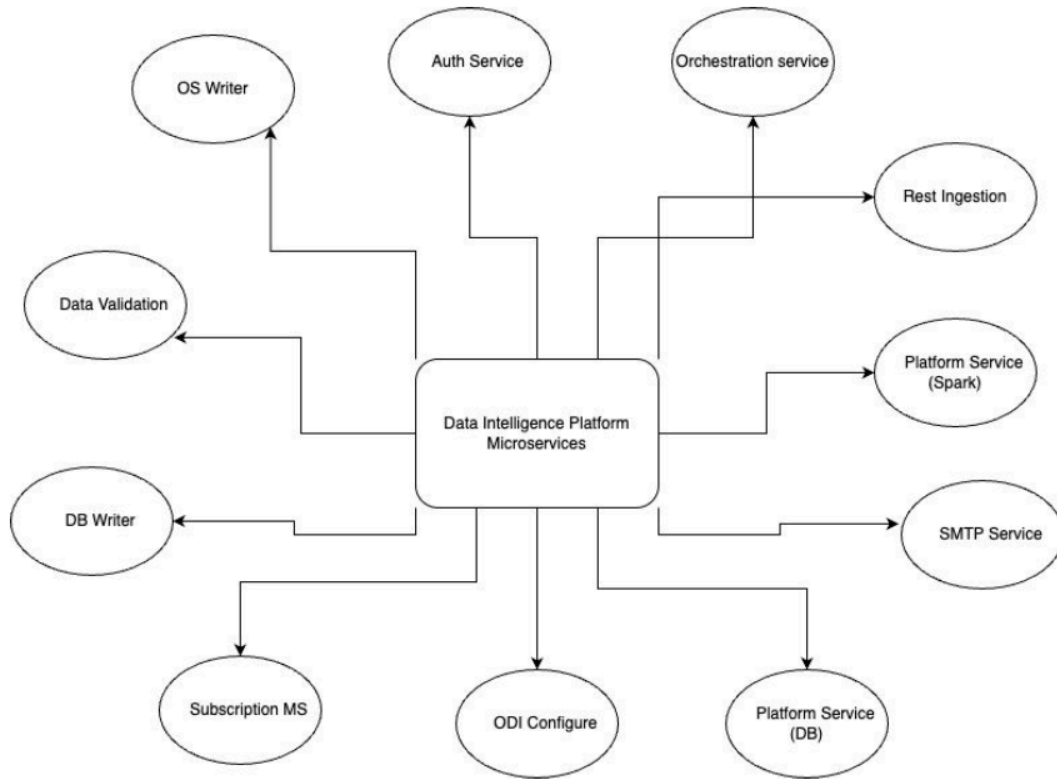


Figure 2.5: DIS Microservices Architecture

2.6.1 Orchestration Service

The **Orchestration Service** is responsible for automating the entire lifecycle of platform components and tenant environments, including initial deployment, configuration, ongoing management, updates, and rollback operations. It coordinates these tasks using template-driven workflows and internal REST APIs, ensuring that all provisioning activities are executed consistently and reliably.

This microservice serves as the central coordinator of the DIS platform by orchestrating interactions among the different services involved in tenant onboarding and lifecycle management. By automating complex operational workflows, it minimizes manual intervention, improves deployment consistency, and enhances the overall reliability of the platform.

The Orchestration Service is the primary focus of this project, as my work was mainly dedicated to designing, implementing, and enhancing its functionality.

2.6.2 Authentication Service

The **Authentication Service** is responsible for managing the secure onboarding and provisioning of tenant identities within the DIS platform. It enforces authentication, authorization, and compliance policies by integrating with Oracle Identity Cloud Service (IDCS) and partner platforms through standardized REST APIs.

This service plays a critical role in ensuring that only authenticated and authorized tenants can access DIS resources. Throughout the provisioning and tenant lifecycle management processes, the Orchestration Service interacts with the Authentication Service to validate partner credentials, provision tenant identities, and manage access permissions. This collaboration guarantees secure communication between services while maintaining compliance with the platform’s security requirements.

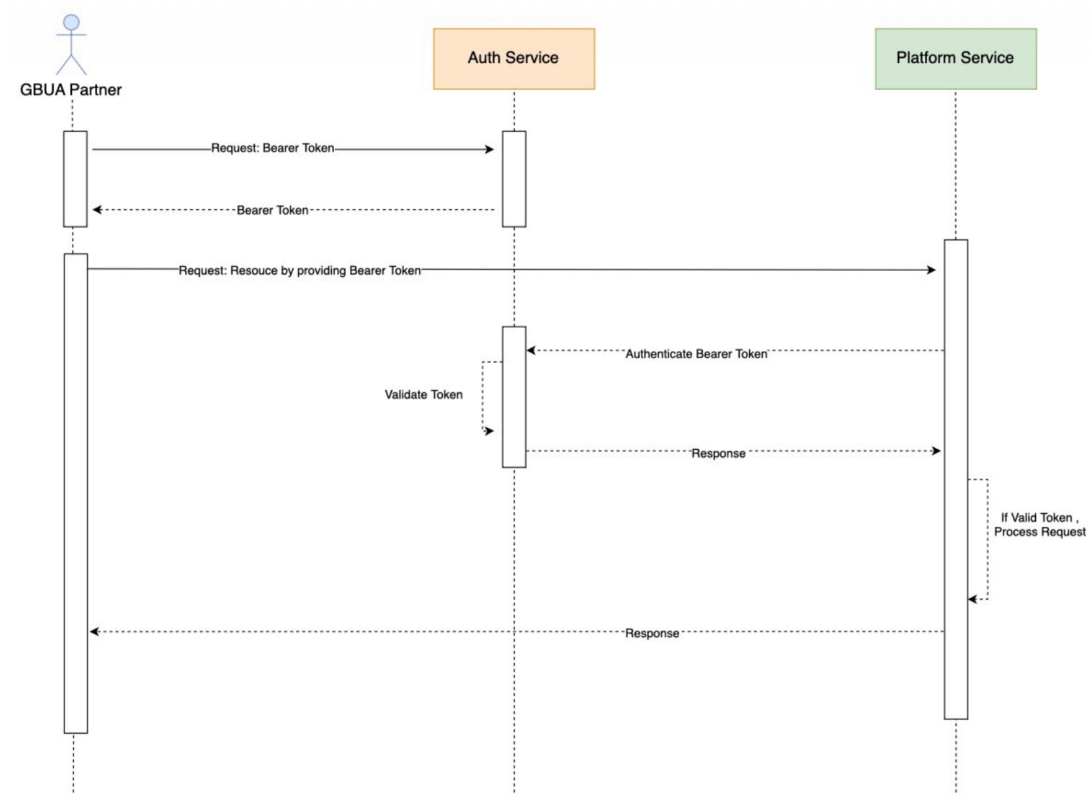


Figure 2.6: Auth Service Sequence Diagram

Conclusion

This chapter positioned DIS within Oracle’s organizational and technical ecosystem. It explained how the platform supports GBUs by providing shared, scalable analytics and machine-learning services across OCI regions. By centralizing these capabilities, DIS reduces duplicated infrastructure, promotes consistent architectural practices, and enables product teams to focus on customer-facing functionality.

Chapter 3

Proposed Solutions and Development

3.1 Introduction

The previous chapter introduced the organizational and technical context of Data Intelligence Services (DIS), as well as the role of Oracle Analytics Cloud (OAC) in delivering analytics capabilities to Oracle’s Global Business Units. This chapter presents the solutions studied and developed during the internship. For each feature, it describes the initial problem, the expected requirements, the proposed architecture, the development approach, and the validation strategy.

3.2 OAC Cross-Region Disaster Recovery Automation

3.2.1 Overview of the Existing Recovery Context

OAC provides cloud-based analytics capabilities such as dashboards, visualizations, datasets, semantic models, and reporting services. During normal operation, each tenant consumes these capabilities from an OAC environment deployed in a primary Oracle Cloud Infrastructure (OCI) region. Although availability mechanisms protect the service from several infrastructure failures, the loss of the complete region requires a separate cross-region recovery strategy.

In the context studied during this internship, cross-region recovery is coordinated by DIS. A standby region is paired with the primary region and must contain, or be able to recreate, the resources required to restore the analytics service. This includes OAC backups, configuration files, deployment artifacts, credentials, network access, and connectivity to external data sources.

Two complementary approaches can be used to prepare OAC in the standby region:

- **Pre-provisioned standby:** an OAC instance is created in the standby region during the initial tenant provisioning. Once verified, it is stopped and remains inactive until a DR event occurs. This option reduces activation time but requires maintaining additional standby resources.

- **On-demand provisioning:** the standby OAC instance is created only after a DR event is declared. This option reduces the resources maintained during normal operation, but the provisioning duration becomes part of the total recovery time.

Figure 3.1 illustrates the first approach, in which the DIS provisioning layer creates analytics environments in both regions.

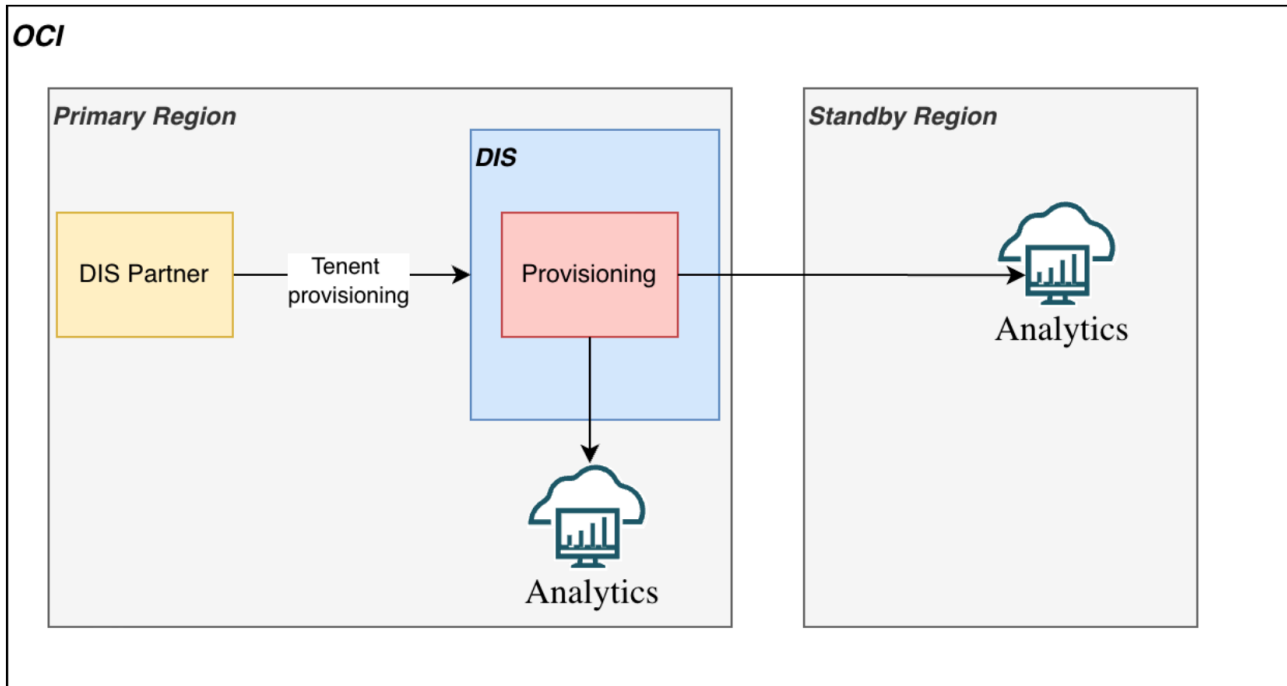


Figure 3.1: OAC provisioning in the primary and standby regions

The automated workflow can support both situations. If a valid standby instance already exists, the system activates or updates it. Otherwise, it provisions a new instance when the DR event is triggered.

3.2.2 Problem Statement

Recovering OAC manually becomes complex when a region contains multiple tenants. Each tenant may have a different configuration and may reach a different state during recovery. A failed operation can also leave partially created resources. For example, an OAC instance may have been created successfully even when the calling service receives a timeout. Restarting the complete workflow without inspecting the existing state could then create duplicate resources or produce inconsistent tenant configuration.

The problem addressed by the feature can therefore be formulated as follows:

How can DIS automate the recovery of multiple OAC tenant environments in a standby region while preserving tenant content, tracking recovery progress, avoiding repeated operations, and safely handling partial failures?

The main difficulties are related to the asynchronous nature of OAC operations, the availability of backups and configuration in the standby region, the management of several tenants, and the coordination of retry and rollback actions. The solution must consequently treat DR as a stateful tenant-level workflow rather than as a single infrastructure command.

3.2.3 Solution Requirements

3.2.3.1 Functional Requirements

The principal functional requirements are presented in Table 3.1.

Table 3.1: Functional requirements of the OAC DR solution

ID	Requirement
FR1	Initiate a DR operation for the tenants affected by a regional outage.
FR2	Determine whether a standby OAC instance already exists for each tenant.
FR3	Start an existing standby instance or provision a new instance when required.
FR4	Restore the appropriate snapshot or BAR backup in the standby environment.
FR5	Recreate the required platform attachment, data connections, and reporting configuration.
FR6	Persist the current status, completed steps, failed step, and error information for each tenant.
FR7	Resume eligible failed tenants without repeating verified completed steps.
FR8	Identify failures that leave partial resources and therefore require rollback.
FR9	Support switchback to the primary region and cleanup of standby resources.

3.2.3.2 Non-Functional Requirements

The solution must also satisfy the following quality requirements:

- **Reliability:** a failure affecting one tenant must not invalidate the recovery of other tenants.
- **Idempotency:** retrying a step must not create duplicate resources or repeat an already verified operation.
- **Consistency:** the persisted tenant status must correspond to the actual state of the resources.
- **Scalability:** the workflow must support several tenants without requiring an independent manual procedure for each one.
- **Security:** credentials, access rights, private connectivity, and network authorization must remain protected in both regions.

- **Auditability:** recovery, retry, rollback, and cleanup operations must be traceable through status information and logs.
- **Testability:** success and failure scenarios must be reproducible in a non-production environment.

3.2.4 Proposed Solution Design

3.2.4.1 General Architecture

The proposed solution follows a warm DR model. During normal operation, the primary region serves the tenants, while the standby region receives the information required for recovery. When a regional failure is confirmed, DIS coordinates the recovery of each tenant through three logical responsibilities:

- a **configuration layer**, which identifies the tenants concerned by the event and initiates recovery;
- an **orchestration layer**, which controls the order of operations and persists their status;
- an **execution layer**, which performs OAC-specific provisioning, restoration, configuration, and cleanup operations.

This separation prevents one service from containing the complete DR logic. It also allows the execution layer to report the technical result of an OAC operation while the orchestration layer decides how the global workflow should continue.

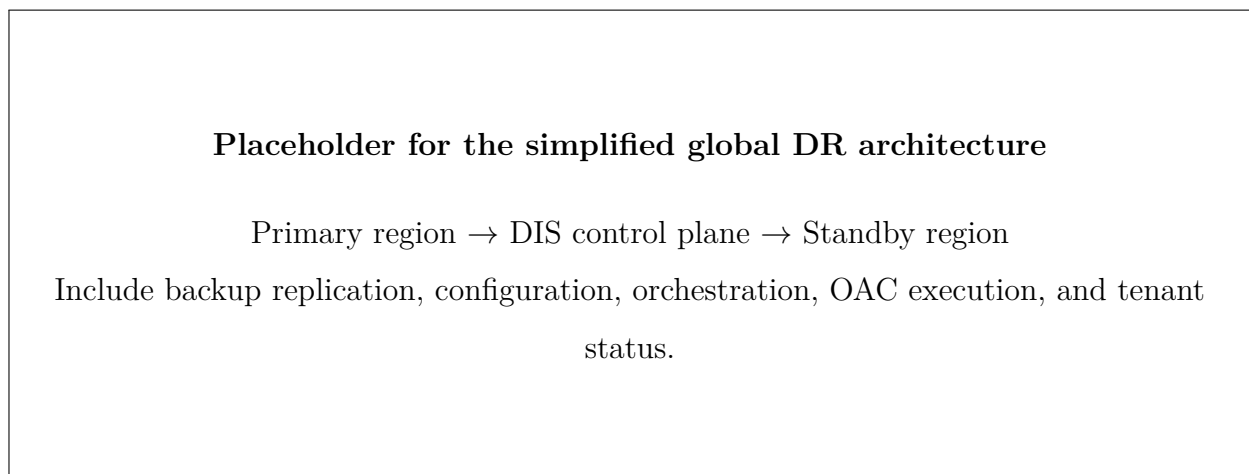


Figure 3.2: Simplified architecture of the OAC cross-region DR solution

The final version of Figure 3.2 should remain intentionally simplified for an external academic reader. Internal URLs, credentials, environment identifiers, and detailed network configuration should not be included.

3.2.4.2 Backup and Cross-Region Preparation

Recovery is possible only if the required tenant information remains available after the loss of the primary region. OAC content is therefore exported regularly as a snapshot or BAR file and stored in Object Storage. The backup and its metadata are then replicated or synchronized to the paired standby region.

The preparation process also covers deployment artifacts, container images, tenant configuration, credentials, access rights, database connectivity, private endpoints, and network allowlists. Storage technologies that do not provide automatic cross-region replication require an explicit synchronization mechanism. These dependencies must be prepared before an incident; provisioning OAC alone would not produce a usable analytics environment.

3.2.4.3 Disaster Recovery Workflow

When a DR event is declared, the recovery workflow is executed for every eligible tenant. Figure 3.3 presents the tenant-level provisioning operation in the standby region.

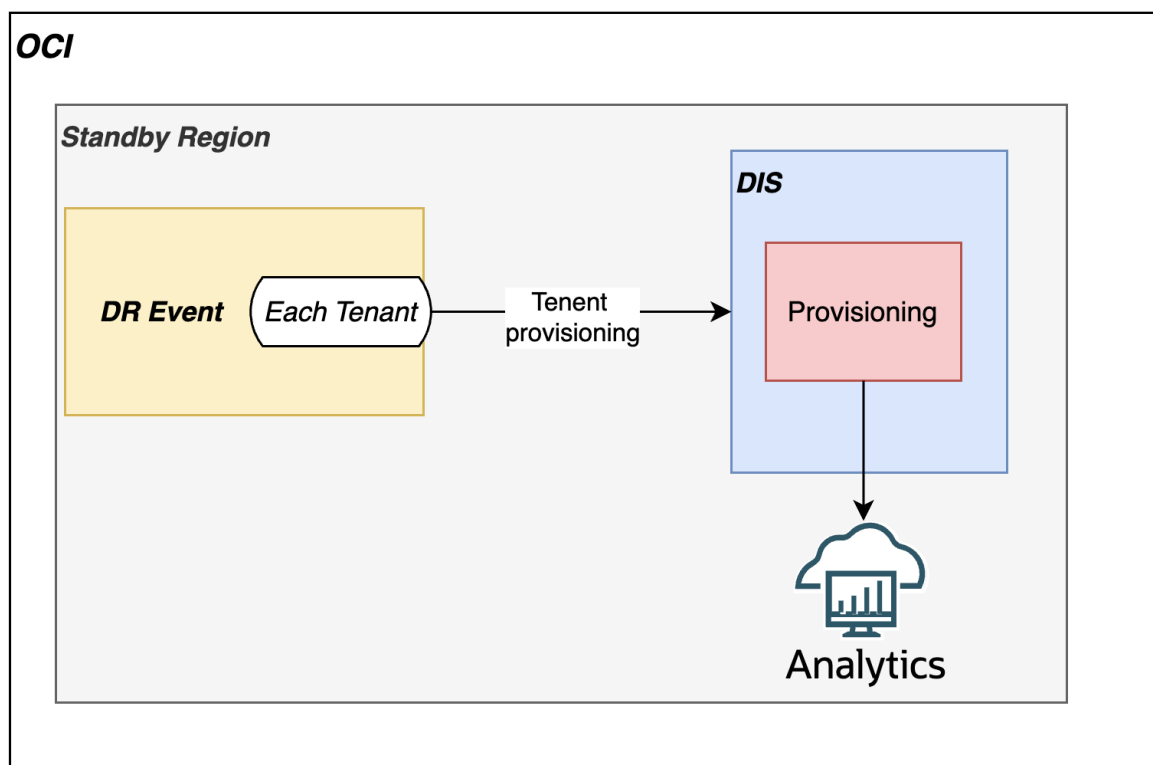


Figure 3.3: Tenant-level OAC provisioning during a DR event

The workflow is composed of the following main steps:

1. The DR operation is initiated for the affected region.
2. The configuration layer identifies the tenants requiring recovery.

3. The orchestration layer creates a workflow and an initial status for each tenant.
4. The system checks whether an OAC instance already exists in the standby region.
5. An existing instance is activated or updated; otherwise, a new instance is provisioned.
6. The platform attachment is created or updated.
7. The selected snapshot or BAR backup is restored.
8. Data Visualization connections and BI Publisher configuration are recreated when required.
9. The standby OAC endpoint and final tenant status are recorded.

The operation is asynchronous. Accepting the initial request does not mean that recovery has completed. Status information is therefore persisted for every tenant using states such as `PENDING`, `IN_PROGRESS`, `SUCCESS`, `FAILED`, and `STOPPED`.

3.2.4.4 Retry and Rollback Strategy

A failed tenant should not systematically restart from the beginning. If OAC provisioning and platform attachment have already completed while snapshot restoration fails, repeating the first two steps could create duplicate resources and increase recovery time. The fix-forward strategy stores the last successful step, verifies the corresponding resource, and resumes execution from the failed operation.

Some failures, however, can leave partial resources that must be removed before another attempt. The OAC execution service therefore sends a rollback decision together with the step status. The value is false by default and becomes true only when a rollback-sensitive operation fails. In the current design, OAC provisioning and platform-attachment creation are classified as rollback-sensitive steps.

The rollback indicator does not perform the deletion itself. It informs the orchestration layer that a simple retry may be unsafe. The orchestration layer can then trigger the appropriate rollback workflow or expose the required recovery action to the operations team.

3.2.4.5 Switchback and Cleanup

The standby region is used until the primary region has been restored and validated. Service is then returned through a controlled switchback, illustrated in Figure 3.4.

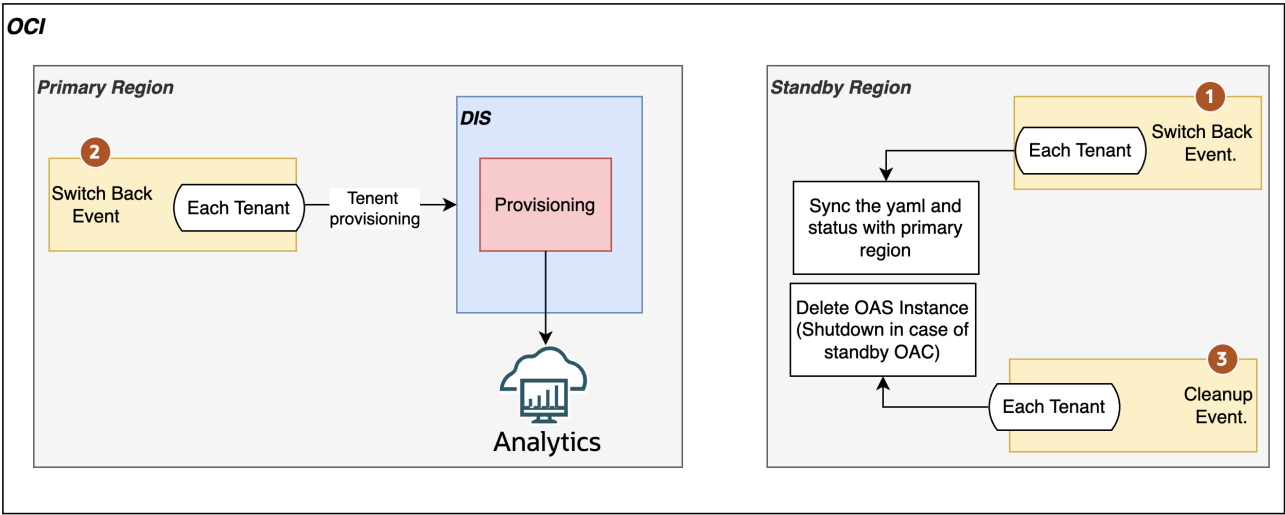


Figure 3.4: Switchback to the primary region and cleanup of standby resources

Switchback consists of three phases. First, configuration, tenant status, and required artifacts are synchronized with the primary region. Second, the OAC environments in the primary region are provisioned or updated and their availability is verified. Finally, temporary resources in the standby region are stopped or deleted. Cleanup must occur only after the primary service and the required backups have been confirmed as available.

3.2.5 Validation

The validation strategy covers the complete recovery lifecycle, from DR initiation to switchback. Table 3.2 summarizes the principal scenarios.

Table 3.2: Principal validation scenarios for OAC cross-region DR

Scenario	Expected result
A DR request is accepted	A tenant workflow is created and its status becomes pending or in progress.
A standby OAC instance already exists	The existing instance is activated or updated without creating a duplicate.
No standby OAC instance exists	A new instance is provisioned in the standby region.
A step fails after previous steps completed	The failure is persisted and verified completed steps are skipped during retry.
Provisioning leaves a partial resource	The tenant is marked as failed and rollback is requested before an unsafe retry.
All recovery steps succeed	The tenant is marked as successful and the standby OAC endpoint is recorded.
The primary region becomes available again	Configuration is synchronized, service is restored in the primary region, and standby cleanup is initiated.

Validation combines unit testing of individual recovery rules, integration testing of communication between services, and end-to-end testing in a non-production region pair. The complete test process must verify actual resource state, backup availability, tenant-level progress, retry, rollback, and switchback.

Some elements remain dependent on the wider platform and operational strategy, including regional-failure detection, DNS redirection, database switchover, and cross-region synchronization of storage systems. These dependencies must be considered when validating the global solution.

3.2.6 Conclusion

The OAC cross-region DR feature transforms regional recovery from a set of manual infrastructure operations into a coordinated tenant-level workflow. The proposed warm DR model combines backup preparation, standby provisioning, status persistence, fix-forward retry, selective rollback, and controlled switchback.

The distinction between fix-forward retry and rollback reduces the risk of repeating completed work or retrying over partially created resources. Together with status tracking and controlled cleanup, it contributes to a more reliable and auditable recovery process. This chapter will be

extended as additional OAC features are addressed during the internship.

Chapter 4

Chapter 4 title

General Conclusion and Perspectives

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Praesent nec dapibus justo. Donec sagittis vulputate ante sed porttitor. Suspendisse sit amet nisl massa. Curabitur nec nisl condimentum, egestas ex vitae, dapibus enim. Etiam iaculis, erat faucibus pellentesque sagittis, nisi justo sollicitudin nibh, et condimentum augue massa non turpis. Proin commodo enim fermentum suscipit condimentum. Maecenas molestie, dui nec vestibulum rhoncus, arcu nisl faucibus neque, a ornare nisi massa ac eros. Aenean id velit sit amet lacus mattis varius. Donec fringilla massa sed nisi eleifend, a aliquet mi tempus. Nunc posuere euismod est, nec tristique augue lobortis non. Sed sodales sem ut metus tempus ullamcorper.

Nullam fermentum id mauris suscipit varius. Quisque tristique tempor fringilla. Nam porta tincidunt orci in consectetur. Suspendisse nec sem nisi. Suspendisse potenti. Sed sodales aliquam libero, at dapibus ligula accumsan non. Quisque congue et urna a consectetur. Nullam maximus venenatis ornare. Donec in luctus urna, vel rhoncus lectus. Etiam lobortis, lacus nec gravida iaculis, magna nulla iaculis leo, quis congue nunc tortor pellentesque lorem. Sed fermentum vulputate dui, ac malesuada eros molestie sit amet. Proin fringilla elit justo, in posuere urna porttitor et. Curabitur dictum justo vitae metus pellentesque, eget feugiat sem feugiat.

Curabitur in mauris eu felis cursus auctor eget ut massa. Curabitur eleifend consectetur magna in ultrices. Etiam ut semper turpis. Morbi sed ipsum nec sem rutrum blandit sit amet vitae felis. Vestibulum vitae hendrerit diam. Aliquam pellentesque est mi, et tempus nisl laoreet in. Maecenas consequat augue a ante congue dignissim. In condimentum erat in volutpat tempus. Mauris vulputate, enim ut dignissim varius, lorem purus tristique sapien, lobortis accumsan dolor lorem quis eros. Duis vel imperdiet metus, non suscipit tortor.

Appendix A

Glossary

Telnet: Telnet is a network protocol that allows users to establish a remote terminal connection to a computer or network device over a network.

SSH (Secure Shell): SSH is a network protocol used for secure remote administration and secure file transfers.

SNMP (Simple Network Management Protocol): SNMP is a network protocol designed for managing and monitoring network devices and systems.

Appendix B

Acronyms

IP	Internet Protocol
CNS	Cloud and Network Services
MN	Mobile Networks
NI	Network Infrastructure
IoT	Internet of Things
NFV	Network Functions Virtualization
SDN	Software-Defined Network
TCP	Transmission Control Protocol
7750 SR	Nokia 7750 Service Router
OSPF	Open short Path First
BGP	Border Gateway Protocol
VPN	Virtual Private Network
MPLS	Multiprotocol Label Switching
EVE-NG	Emulated Virtual Environment - Next Generation
VM	Virtual Machine
SSH	Secure Shell
WinSCP	Windows Secure Copy
FTP	File Transfer Protocol
SSL	Secure Sockets Layer
TLS	Transport Layer Security
FTPS	FTP over SSL/TLS

SFTP	Secure File Transfer Protocol
SCP	Secure Copy Protocol
CLI	Command-line interface
MD-CLI	Model-Driven Command Line
API	Application Programming Interface
re	Regular Expression
IGP	Internal Gateway Protocol
IS-IS	Intermediate-System Intermediate-System
SPF	Shortest Path First
SF	Switch Fabric
CPM	Control Processing Module
IOM	Input/Output Modules
MDA	Media Dependent Adapters
QoS	Quality of Service
IPsec	Internet Protocol Security
VSR	Virtual Services Router
DDoS	Distributed Denial-of-Service
SROS	Service Router Operating System
NETCONF	Network Configuration Protocol
YANG	Yet Another Next Generation
MP-BGP	Multiprotocol Border Gateway Protocol
IPv6	Internet Protocol Version 6
AS	Autonomous Systems
iBGP	Internal Border Gateway Protocol
eBGP	External Border Gateway Protocol
LAG	Link Aggregation Group
VPN	Virtual Private Routed Network
SNMP	Simple Network Management Protocol
Telnet	Telecommunication Network
YAML	Yet Another Markup Language
XML	eXtensible Markup Language

Appendix C

Some subject you want to expand on

You can add more appendices depending on the subjects.

```
1
2 [root@host ~]# You can change the language and caption.
3
4
```

Listing C.1: Bash example

```
1
2 # Solve the quadratic equation ax**2 + bx + c = 0
3
4 # import complex math module
5 import cmath
6
7 a = 1
8 b = 5
9 c = 6
10
11 # calculate the discriminant
12 d = (b**2) - (4*a*c)
13
14 # find two solutions
15 sol1 = (-b-cmath.sqrt(d))/(2*a)
16 sol2 = (-b+cmath.sqrt(d))/(2*a)
17
18 print('The solution are {0} and {1}'.format(sol1,sol2))
19
20
```

Listing C.2: Python example

Column1	Column2
Element11	Element21
Element12	Element22
Element13	Element23

Table C.1: Table Example

Table C.2: Long table Example

Cell	Description
Element11	Element21
Element12	Element22
Element13	Element23
Element14	Element24
Element15	Element25
Element16	Element26
Element17	Element27
Element18	Element28
Element19	Element29
Element110	Element210
Element111	Element211
Element112	Element212
Element113	Element213
Element114	Element214

Bibliography

[1] Author name, Book name.

[2] *Title 1*, **Title 2**